Euclid's Elements — Book I
Proposition 41

Formal Reconstruction — Technical Documentation

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If a parallelogram has the same base with a triangle and is in the same parallels, then the parallelogram is double the triangle.

1 Introduction

Proposition I.41 begins a new use of the area block rather than a new area foundation. The preceding propositions I.37–I.40 compare triangles. I.41 combines the same-base theorem I.37 with the diagonal theorem I.34 and concludes that a parallelogram is "double" a triangle on the same base and between the same parallels.

The present formalization keeps this statement entirely synthetic. It does not introduce a numerical area function and it does not interpret "double" as multiplication by the real number 2. Instead the conclusion is expressed in the existing scissors language: the formal parallelogram term is equicomplementable with the formal sum of two copies of the triangle term. Thus the relevant target has the form

$$P \approx (T + T),$$

where $\approx$ abbreviates the project's term-level

```
HilbertScissorsEquicomplementable
```

relation.

I.41 is therefore a forward content theorem. No content relation is inverted, no proper-part comparison is required, and no positivity, cancellation, or whole-greater-than-part principle is introduced. This sharply separates I.41 from the converse propositions I.39 and I.40.

The reconstruction also exposed one missing reusable law of the scissors calculus. Equicomplementability must be compatible with formal addition. The initial proposition-local "doubling" helper was consequently replaced by the general theorem

```
equicomplementable_add
```

in `HilbertScissors.lean`. The full project rebuild remained clean, and the same reusable theorem now also replaces the former I.41-specific helper in Proposition I.47.

2 The Classical Configuration

Let $ABCD$ be a parallelogram and let EBC be a triangle with the same base BC. Both figures lie between the same two parallel carriers. In Euclid's notation the upper carrier passes through A and E, so

$$AE \parallel BC.$$

The parallelogram has the same lower base BC, and its upper side AD lies on the same upper carrier in the classical picture.

The public Lean theorem encodes this configuration economically by the two hypotheses

```
(hParallelogram : IsParallelogram Geo A B C D)
(hAE_BC : Geo.Parallel A E B C)
```

There is no additional hypothesis asserting explicitly that A, D, E are collinear. The proof does not need such a witness. The parallelogram structure is used by I.34, while the parallel pair AE, BC is used independently by I.37. This is an example where the formal statement contains exactly the data needed by the proof rather than reproducing every incidence relation visible in the classical diagram.

Figure 1 shows both the classical configuration and the two content comparisons used by the Lean proof.

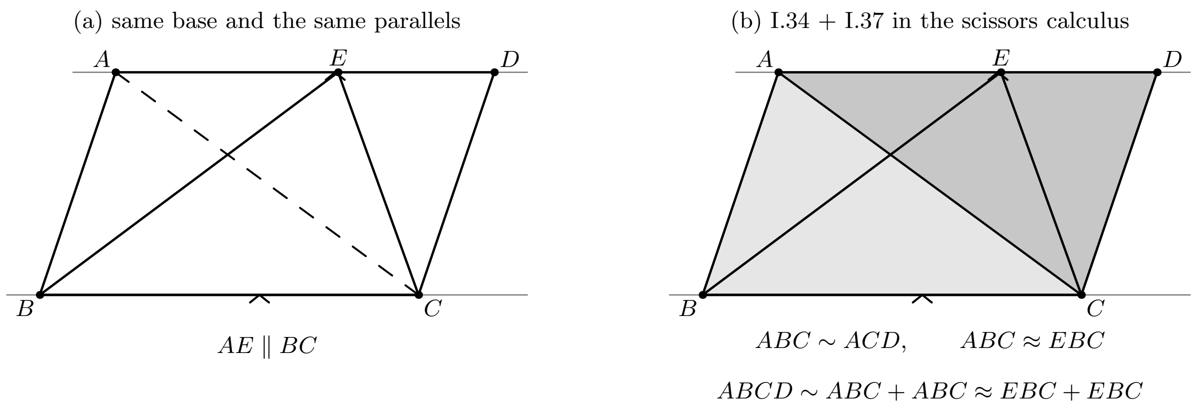

Figure 1: The configuration and proof structure of I.41. (a) The parallelogram $ABCD$ and triangle EBC share the base BC and lie between the parallel carriers through AE and BC; the diagonal AC is the diagonal used by I.34. (b) I.34 makes the two diagonal triangles scissors-equivalent, so the parallelogram term can be represented as two copies of ABC. I.37 gives equal content for ABC and EBC. The reusable addition theorem then transports this relation to the two formal sums.

3 Classical Proof

Euclid's proof is one of the shortest in the area block.

Join A to C. Proposition I.37 applies to the triangles ABC and EBC: they stand on the same base BC and lie in the same parallels AE and BC. Hence the two triangles are equal in Euclid's content sense.

Next apply Proposition I.34 to the parallelogram. Its diagonal AC bisects the parallelogram, so the parallelogram is double the triangle ABC. Since ABC is equal to EBC, the parallelogram is also double the triangle EBC.

The classical dependency graph is therefore

$$\begin{array}{c}
ABCD \text{ parallelogram}, \quad AE \parallel BC \\
\Downarrow \\
\triangle ABC = \triangle EBC \qquad \text{by I.37} \\
\Downarrow \\
ABCD = 2\triangle ABC \qquad \text{by I.34} \\
\Downarrow \\
ABCD = 2\triangle EBC.
\end{array}$$

Joyce's text makes exactly this sequence explicit: join AC, use I.37 for the same-base triangles, and use I.34 for the diagonal bisection of the parallelogram. Unlike I.39 and I.40, there is no reductio and no proper-part argument.

4 Doubling in the Scissors Calculus

The word "double" needs a precise synthetic interpretation in the current library. The type

```
HilbertScissorsTerm
```

is a formal multiset sum of triangle terms. Thus the term-level double of a triangle T is simply

$$T + T.$$

No scalar multiplication is introduced.

The crucial reusable statement is

```
theorem equicomplementable_add
    {P1 Q1 P2 Q2 : HilbertScissorsTerm Geo}
    (h1 : HilbertScissorsEquicomplementable Geo P1 Q1)
    (h2 : HilbertScissorsEquicomplementable Geo P2 Q2) :
    HilbertScissorsEquicomplementable Geo
      (P1 + P2) (Q1 + Q2)
```

Conceptually, if $P_1 \approx Q_1$ and $P_2 \approx Q_2$, then

$$P_1 + P_2 \approx Q_1 + Q_2.$$

I.41 uses the same relation twice, obtaining

$$ABC + ABC \approx EBC + EBC.$$

This theorem belongs to `HilbertScissors.lean`, not to `Proposition41.lean`. Its proof is pure algebra of the existing scissors relation: the complements for the two hypotheses are added, the two whole-term relations are combined with `HilbertScissorsEq.add`, and associativity and commutativity of multiset addition rearrange the resulting terms.

This is stronger architecturally than an I.41-specific “double” lemma. The law is closed under arbitrary pairs of equicomplementable terms and can be reused whenever a later proposition adds equal-content components.

5 Proof Architecture of the Reconstruction

The Lean proof is short enough to be displayed as four conceptual stages.

5.1 Step 1: replace the second diagonal triangle by the first

Proposition I.34 supplies

```
euclid_proposition_34_diagonal
```

with result

$$ABC \cong CDA.$$

The theorem

```
scissors_congruent
```

turns this triangle congruence into a scissors equality. A cyclic permutation of the triangle representation identifies CDA with ACD, giving

$$ABC \sim ACD,$$

where $\sim$ denotes `HilbertScissorsEq`.

The formal parallelogram term is defined by the diagonal AC:

$$\mathrm{Par}(ABCD) = ABC + ACD.$$

Adding reflexivity on the first triangle to the reverse of the relation above yields

$$\mathrm{Par}(ABCD) \sim ABC + ABC.$$

This is the formal counterpart of the classical statement that the diagonal bisects the parallelogram.

5.2 Step 2: apply Proposition I.37 on the common base

The hypothesis

$$AE \parallel BC$$

is already in exactly the form expected by Proposition I.37. The call

```
euclid_proposition_37 Geo A B C E hAE_BC
```

returns

$$ABC \approx EBC.$$

No new carrier construction, SameSide witness, midpoint construction, or order case split appears in I.41 itself. All geometric normalization required by the same-base theorem is encapsulated in the already proved I.37.

5.3 Step 3: add the equal-content relation to itself

The new reusable theorem is applied twice to the same hypothesis:

```
equicomplementable_add
  Geo
  hABC_EBC
  hABC_EBC
```

This gives

$$ABC + ABC \approx EBC + EBC.$$

This is the precise term-level operation corresponding to Euclid's use of "double."

5.4 Step 4: transport from the doubled triangle to the parallelogram

The final call

```
equicomplementable_transport
```

uses the scissors equality

$$\mathrm{Par}(ABCD) \sim ABC + ABC$$

on the left and reflexivity on $EBC + EBC$ on the right. It therefore transports the doubled I.37 conclusion to the public target

$$\mathrm{Par}(ABCD) \approx EBC + EBC.$$

The complete Lean DAG is

$$\begin{array}{c}
\texttt{IsParallelogram}(A,B,C,D) \quad + \quad AE \parallel BC \\
\Downarrow \quad \text{I.34} \\
ABC \cong CDA \\
\Downarrow \quad \texttt{scissors_congruent} \\
\mathrm{Par}(ABCD) \sim ABC + ABC \\
AE \parallel BC \\
\Downarrow \quad \text{I.37} \\
ABC \approx EBC \\
\Downarrow \quad \texttt{equicomplementable_add} \\
ABC + ABC \approx EBC + EBC \\
\Downarrow \quad \texttt{equicomplementable_transport} \\
\mathrm{Par}(ABCD) \approx EBC + EBC \\
\Downarrow \\
\texttt{euclid_proposition_41}.
\end{array}$$

6 Lean Formalization

The current proposition file imports exactly the two earlier propositions used by the mathematical proof:

```
import CGJteamLab.Proposition34
import CGJteamLab.Proposition37
```

The public theorem is:

```
theorem euclid_proposition_41
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C D E : Geo.Point)
    (hParallelogram : IsParallelogram Geo A B C D)
    (hAE_BC : Geo.Parallel A E B C) :
    HilbertScissorsEquicomplementable Geo
      (hilbertParallelogramTerm Geo A B C D)
      (hilbertScissorsTriangle Geo E B C +
       hilbertScissorsTriangle Geo E B C) := by
  ...
```

The ellipsis is documentation notation only. The repository theorem is fully proved and the full project build is clean.

The mathematically significant declarations called directly by the proof are:

```
euclid_proposition_34_diagonal
scissors_congruent
scissors_triangle_cycle
euclid_proposition_37
equicomplementable_add
equicomplementable_transport
```

The remaining operations are reflexivity, symmetry, and formal addition in `HilbertScissorsEq`. There are no proposition-local helper theorems in the final I.41 file.

7 Relationship with Earlier Propositions

7.1 Proposition I.34

I.34 supplies the diagonal congruence of the parallelogram. In the classical proof this is stated as bisection of the parallelogram by its diagonal. In the formal proof the stronger triangle-congruence result is converted directly into scissors equality and then used to rewrite

$$ABC + ACD$$

as

$$ABC + ABC.$$

Thus I.34 is the representation bridge from the geometric parallelogram to the formal doubled triangle.

7.2 Proposition I.37

I.37 is the direct content theorem. It states that triangles on the same base and between the same parallels are equicomplementable. With the substitution $D := E$, its public hypothesis is exactly the I.41 hypothesis $AE \parallel BC$. The result is therefore available without any new geometric construction.

I.37 is also the reason I.41 needs no halving or cancellation theorem. The formal proof proceeds in the forward direction

$$ABC \approx EBC \quad \Longrightarrow \quad ABC + ABC \approx EBC + EBC,$$

not in the converse direction from doubled figures back to their halves.

7.3 Propositions I.35–I.40

I.35–I.38 established the forward equal-content machinery; I.39 and I.40 then showed where converse reasoning requires strict content information. I.41 returns to the forward direction. Consequently it uses only equivalence transport and addition in the scissors calculus. The positivity issue isolated for I.39 is not needed here.

This position in the Book I sequence is important: I.41 is not another test of proper-part semantics. It is a test of whether the already established content relation behaves compositionally under addition.

8 Hilbert and Book Zero Components Used

8.1 Triangle congruence and parallelogram infrastructure

The direct geometric input from I.34 ultimately rests on the established Hilbert congruence and parallelogram theory. I.41 itself does not reopen SSS, angle transport, or diagonal incidence. It consumes the packaged theorem

```
euclid_proposition_34_diagonal
```

as a black box.

8.2 Same-base triangle content

All of the substantial geometry behind equal-content triangles on a common base is encapsulated by I.37. Its Hilbertian reconstruction uses the midpoint and parallelogram machinery developed earlier in the project, but none of those witnesses is reconstructed in I.41.

8.3 Scissors representation

Three levels must remain distinct:

- `hilbertParallelogramTerm` is the chosen formal triangulation of a parallelogram along AC;
- `HilbertScissorsEq` is the stronger scissors-equivalence relation;
- `HilbertScissorsEquicomplementable` is the equal-content relation used in the public theorem.

I.34 first produces the stronger relation; I.37 produces the weaker content relation; I.41 combines them by addition and transport.

No new Book Zero theorem is introduced. The only new reusable declaration in the scissors layer is

```
equicomplementable_add
```

9 Source Audit

9.1 Euclid and Joyce

The classical source gives exactly the short proof reflected in the present architecture: join AC, apply I.37 to ABC and EBC, then use I.34 to say that the parallelogram is double the diagonal triangle. Joyce also notes that I.41 generalizes the diagonal statement I.34 and that an equal-base version would be slightly more general.

9.2 Hartshorne

Hartshorne's Section 22 is the main semantic control for the area block. His classification explicitly places I.41–I.45 among the remaining Book I results that are valid for equal content. This is consistent with the present proof: I.41 never uses equal content as a hypothesis to recover metric or incidence information and therefore needs no new property (Z)-style strictness principle.

9.3 Hilbert, Borsuk–Szmielew, Greenberg, and Beeson

For I.41 these sources play their established control roles rather than supplying a new proposition-specific theorem. Hilbert remains the architectural source for the separation between congruence, equidecomposition, and equicomplementability. Borsuk–Szmielew and Greenberg control the developed incidence/congruence/parallel background. The Beeson corpus remains a check on formal side conditions, but no Tarski-style local construction is imported into the final Lean proof.

The current Lean source is authoritative for the exact formal claim. In particular, the documentation does not strengthen its conclusion to numerical area equality or to an unconditional geometric dissection statement.

10 Formalization Notes

10.1 Geometry

The public theorem exposes only two geometric inputs:

- the parallelogram structure of $ABCD$;
- the strict parallel relation $AE \parallel BC$.

There are no explicit Between, SameRay, SameSide, OppositeSide, Pasch, or extension hypotheses in I.41. Any lower-level nondegeneracy needed by I.34 or I.37 is handled inside those earlier theorems.

The absence of an explicit A, D, E collinearity hypothesis is intentional. It is not needed by the actual proof DAG: I.34 consumes the parallelogram, and I.37 consumes $AE \parallel BC$.

10.2 Content

The content direction is purely forward:

$$ABC \approx EBC \quad \Longrightarrow \quad ABC + ABC \approx EBC + EBC.$$

Accordingly there is no subtraction, cancellation, positivity, strict content order, or whole-greater-than-part step. The only new content fact is the closure of equicomplementability under formal addition.

10.3 Representation

The diagonal theorem returns the triangle orientation $ABC \cong CDA$, while the parallelogram term is written as $ABC + ACD$. Two applications of

```
scissors_triangle_cycle
```

identify the cyclic presentations. This is representation transport only; it contains no new geometric argument.

The final statement should therefore be read literally at the term level:

$$\text{parallelogram term} \approx \text{triangle term} + \text{triangle term}.$$

It is the current scissors representation of Euclid's "double," not a claim about a real-valued area function.

10.4 Axiomatic status

The public theorem assumes

```
HilbertEuclideanPlane Geo
```

and is therefore Euclidean in the current hierarchy. I.41 introduces no new parallel construction and does not call parallel uniqueness directly; its Euclidean dependence is inherited through the already proved I.34/I.37 layer, most importantly the same-parallels content theorem I.37.

No continuity axiom is introduced. No new geometric or content axiom is added by Proposition I.41.

11 Dependency Summary

The actual formal dependency chain is

$$\begin{gathered}
\text{Hilbert incidence, congruence, order, and Euclidean parallel layer} \\
\Downarrow \\
\text{HilbertInterface / Book Zero / HilbertScissors} \\
\Downarrow \\
\text{Proposition I.34 diagonal congruence} \quad + \quad \text{Proposition I.37 same-base content} \\
\Downarrow \\
\texttt{scissors_congruent} \quad + \quad \texttt{equicomplementable_add} \\
\Downarrow \\
\mathrm{Par}(ABCD) \sim ABC + ABC \quad + \quad ABC + ABC \approx EBC + EBC \\
\Downarrow \\
\texttt{equicomplementable_transport} \\
\Downarrow \\
\texttt{euclid_proposition_41}.
\end{gathered}$$

The historical and formal DAGs are unusually close:

$$\begin{aligned}
\text{Euclid} &: \ \text{I.37} + \text{I.34} \longrightarrow \text{I.41}, \\
\text{Lean} &: \ \text{I.37} + \text{I.34} + \text{scissors addition/transport} \longrightarrow \text{I.41}.
\end{aligned}$$

The extra Lean layer is representational rather than geometric.

New geometric axiom in I.41:	no
New content axiom in I.41:	no
Inherited strict-content/positivity axiom:	no
New proposition-local helper:	no
New reusable declaration:	yes — `equicomplementable_add`
Reusable refactor required:	yes — helper moved to HilbertScissors
New Interface/Scissors theorem:	yes
New Book Zero theorem:	no
New numerical area or polygon semantics:	no
Public theorem compiles:	yes
Full `lake build`:	clean
`sorry` in `Proposition41.lean`:	no

12 Conclusion

Proposition I.41 is a compact confirmation that the area architecture developed since I.35 is compositional. Its geometry is already contained in two earlier results: I.34 identifies the two diagonal halves of a parallelogram, and I.37 compares triangles on the same base and between the same parallels. The new formal ingredient is not another geometric construction but the algebraic fact that equicomplementability is preserved by formal addition.

That observation led directly to the reusable theorem

```
equicomplementable_add
```

in `HilbertScissors.lean`. Once this theorem is available, the I.41 proof becomes almost the same DAG as Euclid's: replace the second diagonal triangle by the first, apply I.37, double the content relation by addition, and transport back to the parallelogram term.

The proposition therefore introduces no new axiom, no numerical area, no strict content order, and no new polygon semantics. The full project rebuild is clean, and the general addition theorem is already reusable beyond I.41.